

REFINERY CENTRIFUGAL PUMP MAINTENANCE MANUAL

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Asset Class: Centrifugal Pump

Purpose: Reliability-Centered Maintenance Program

1. Purpose

This document defines preventive, predictive, corrective, and reliability-centered maintenance practices for refinery centrifugal pumps.

The objective is to:

- Maximize equipment availability
 - Minimize unplanned downtime
 - Extend asset life
 - Improve safety
 - Support AI-driven maintenance decisions
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2. Maintenance Philosophy

Pump failures rarely occur suddenly.

Most failures develop gradually and produce measurable telemetry changes long before breakdown occurs.

The maintenance strategy is based on:

Preventive Maintenance

Scheduled maintenance performed at predefined intervals.

Examples:

- Lubrication replacement

- Seal inspection
 - Alignment verification
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Predictive Maintenance

Maintenance triggered by equipment condition.

Examples:

- Increasing vibration
 - Increasing bearing temperature
 - Rising motor current
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Corrective Maintenance

Maintenance performed after a defect is identified.

Examples:

- Bearing replacement
 - Seal replacement
 - Shaft repair
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Reliability-Centered Maintenance

Maintenance decisions based on:

- Failure probability
 - Remaining useful life
 - Operational criticality
 - Production impact
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3. Daily Inspection Program

Daily inspections provide the earliest opportunity to identify abnormal conditions.

Bearing Temperature Review

Telemetry:

bearing_temp

Inspection Frequency:

Daily

Normal:

50°C – 75°C

Inspection Actions:

- Verify temperature trend
- Compare with previous shift
- Identify abnormal increases

Warning Signs:

- Continuous upward trend
- Temperature rise >5°C per day

Possible Causes:

- Lubrication degradation
- Bearing wear
- Misalignment

Vibration Review

Telemetry:

vibration

Inspection Frequency:

Daily

Normal:

0 – 4 mm/s

Inspection Actions:

- Review vibration trend
- Compare with baseline
- Listen for abnormal noise

Warning Signs:

- Increasing trend
- Sudden spikes

Possible Causes:

- Bearing wear
 - Cavitation
 - Rotor imbalance
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Flow Rate Verification

Telemetry:

flow_rate

Inspection Frequency:

Daily

Purpose:

Verify pump performance.

Abnormal Conditions:

- Reduced throughput
- Flow instability

Potential Causes:

- Suction blockage
 - Cavitation
 - Seal leakage
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Seal Leakage Inspection

Telemetry:

seal_leakage_rate

Inspection Frequency:

Daily

Inspection Method:

- Visual inspection
- Leakage collection measurement

Acceptable Leakage:

0 – 0.1 L/hr

Action Required:

0.5 L/hr

4. Weekly Inspection Program

Weekly inspections focus on developing degradation mechanisms.

Lubrication System Inspection

Telemetry:

lubrication_level

Inspection Tasks:

- Verify lubricant level
- Check lubricant contamination
- Verify lubricant color
- Inspect breather condition

Indicators of Degradation:

- Metal particles

- Dark coloration
- Water contamination

Consequences:

- Bearing overheating
- Increased vibration
- Reduced RUL

Motor Current Assessment

Telemetry:

motor_current

Inspection Frequency:

Weekly

Normal:

80 – 120 A

Analysis:

Compare current against:

- Historical baseline
- Similar pumps
- Load conditions

Increasing Current May Indicate:

- Bearing drag
- Mechanical resistance
- Rotor problems

Pressure Verification

Telemetry:

suction_pressure

discharge_pressure

Inspection Objectives:

- Verify hydraulic performance
 - Detect restrictions
 - Detect leakage
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5. Monthly Inspection Program

Monthly inspections provide detailed health assessments.

Bearing Inspection

Procedure:

Step 1:

Review vibration history

Step 2:

Review temperature history

Step 3:

Inspect bearing housing

Step 4:

Inspect lubricant condition

Step 5:

Evaluate replacement need

Seal Inspection

Procedure:

Step 1:

Measure leakage

Step 2:

Inspect seal faces

Step 3:

Inspect shaft sleeve

Step 4:

Evaluate wear condition

Alignment Verification

Misalignment creates:

- Increased vibration
- Increased bearing load
- Reduced efficiency

Verification Methods:

- Laser alignment
- Dial indicator alignment

Acceptable Offset:

According to site standards.

6. Quarterly Maintenance Program

Quarterly activities focus on predictive diagnostics.

Vibration Analysis

Required Data:

- Overall vibration
- Frequency spectrum

Objectives:

Identify:

- Bearing defects
 - Imbalance
 - Misalignment
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Thermography

Equipment:

Infrared camera

Targets:

- Bearings
- Motor
- Couplings

Benefits:

Detect hot spots before failure.

Performance Assessment

Evaluate:

- Flow rate
- Pressure
- Efficiency

Compare against design performance.

7. Annual Overhaul Program

The annual overhaul provides a complete mechanical assessment.

Scope

Includes:

- Bearing replacement
 - Seal replacement
 - Shaft inspection
 - Impeller inspection
 - Coupling inspection
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Bearing Inspection

Inspect:

- Raceways
- Rolling elements
- Lubrication condition

Reject Criteria:

- Pitting
 - Spalling
 - Excessive wear
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Impeller Inspection

Inspect:

- Erosion
- Corrosion
- Cavitation damage

Impact of Damage:

- Reduced flow
 - Reduced efficiency
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Shaft Inspection

Inspect:

- Runout

- Surface damage
 - Sleeve wear
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8. Reliability Indicators

The following telemetry parameters have the strongest relationship with pump health.

Bearing Temperature

Primary Health Indicator

Strong Predictor of:

- Bearing failure
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Vibration

Primary Reliability Indicator

Strong Predictor of:

- Bearing wear
 - Cavitation
 - Misalignment
-

Motor Current

Secondary Reliability Indicator

Strong Predictor of:

- Mechanical resistance
 - Friction
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Seal Leakage Rate

Primary Seal Health Indicator

Strong Predictor of:

- Seal failure
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9. AI Copilot Maintenance Logic

Condition:

Failure Probability > 70%

Action:

Generate maintenance notification.

Condition:

Failure Probability > 85%

RUL < 30 Days

Action:

Schedule maintenance.

Condition:

Failure Probability > 90%

RUL < 15 Days

Action:

Reliability engineer review required.

Condition:

Failure Probability > 95%

RUL < 7 Days

Action:

Immediate maintenance planning.

10. Maintenance Decision Matrix

Bearing Temperature ↑

Vibration ↑

Motor Current ↑

Diagnosis:

Bearing Failure

Action:

Bearing replacement

Seal Leakage ↑

Discharge Pressure ↓

Flow Rate ↓

Diagnosis:

Seal Failure

Action:

Seal replacement

Suction Pressure ↓

Flow Rate ↓

Vibration ↑

Diagnosis:

Cavitation

Action:

Investigate suction system

11. Summary

Effective pump maintenance requires continuous monitoring of:

- Bearing Temperature
- Motor Temperature
- Vibration
- Flow Rate
- Suction Pressure
- Discharge Pressure
- Motor Current
- Lubrication Level
- Seal Leakage Rate

Maintenance decisions should combine:

- Telemetry
- Failure Probability
- Failure Mode
- Remaining Useful Life

to maximize reliability and minimize downtime.